

Theory Lectures on Manufacturing Process (NMES 101) Casting / Foundry Technology

L1: Introduction



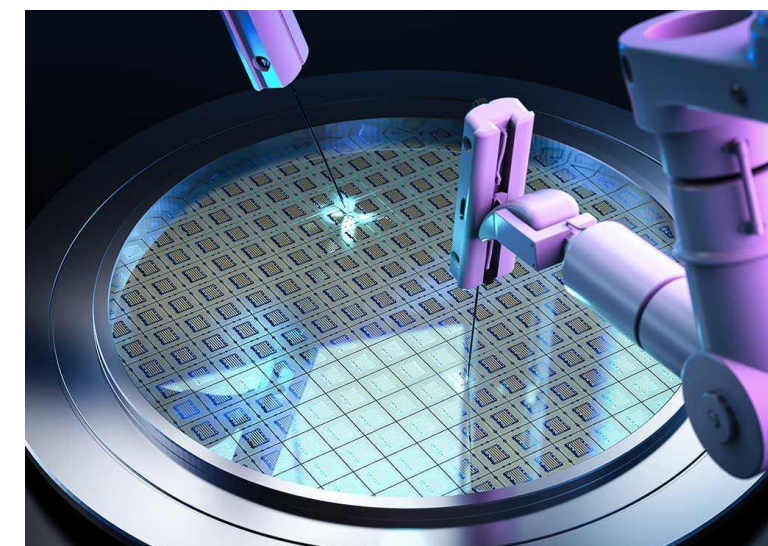
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Content prepared for theory lecture for
1st Year BTech level students for common
course on Manufacturing Processes.
Each lecture duration ≈ 45 min



What is a product?

A product is a tangible or intangible item that fulfills the need of the customer.



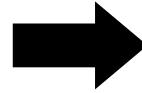
If the product is not naturally occurring, then it needs to be manufactured.

What is meant by manufacturing?

3



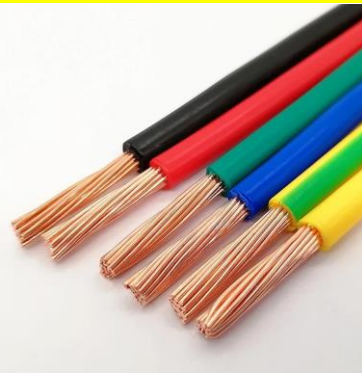
Mining
Excavation



Mechanical
Manufacturing



Every physical product needs to be **manufactured!** The product can vary from simple pen, vehicle to jewellery to supercomputer.



Manufacturing refers to a cluster of value-addition processes that are used to convert raw materials into high-utility and high-value products having intended shape, size, finish, and functionality.

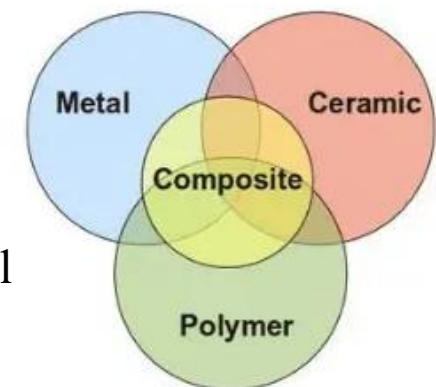
Countries with high manufacturing output (2021)

- | | |
|----------------------------------|------------------------------|
| 1. China (USD 4,975,614) | 4. Germany (USD 751,338) |
| 2. United States (USD 2,497,131) | 5. South Korea (USD 456,600) |
| 3. Japan (USD 1,025,092) | 6. India (USD 446,504) |

Engineering materials

1. Metals
2. Ceramics
3. Polymer

A **composite** is a material made from two or more different materials



Difference between manufacturing & production

4

Think about a manufacturing industry – say car making industry

From where they will get the raw materials? **MRP planning**

When they will order the raw materials? **Forecasting**

Where they will store the raw materials? **Inventory control**

Which product they need to make – specification? **Designing**

How they will convert the raw material into the product? **Manufacturing**

In which sequence the product will be fed? **Scheduling**

How they will maintain the machines? **Maintenance**

How they will control the quality of the product? **Quality control**

How they will join different parts to make the final product? **Assembly**

How they will package the product? **Packaging**

How the price will be valued? **Strategic cost planner**

How the product will be delivered in timely manner? **Logistics**

Who will collect customer feedback? **Customer service**

Finally, who will manage all these things? **Management**

Production $\supset$ Manufacturing



Production

A system

It comprises all relevant fields of the fabrication process

Involves following steps:
Forecasting, management, designing, purchasing, scheduling, manufacturing, maintenance, inspection, quality control, packaging, sales, customer service, etc.

Manufacturing

A step

It actually converts raw material or scrap into desired product

Involves following processes:
Casting and molding, forming, joining, machining, heat treatment, coating and coloring, rapid prototyping, powder metallurgy, etc.



7 major groups of manufacturing

How to select appropriate manufacturing process? What are the key factors to judge applicability and suitability of a process?

Product shape



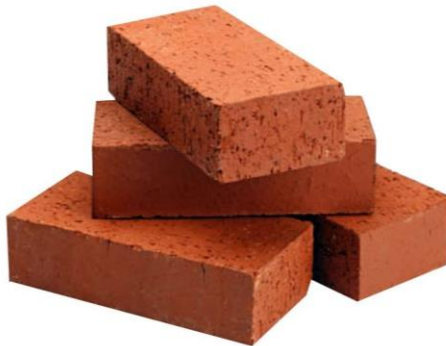
Product size



Product material



Surface finish quality (roughness level)



Batch volume (number of products to be made)



Cost of the product



Accordingly, seven different manufacturing processes also exist to fulfil the requirement in best possible way

Casting

Forming

Joining

Machining

Powder Metallurgy

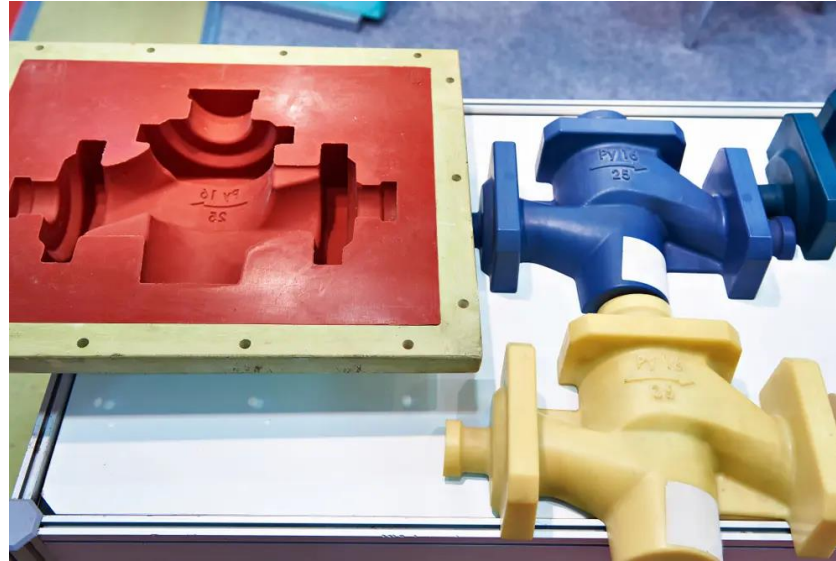
Additive Manufacturing

Surface Engineering

Casting – One type of manufacturing process



Melt the raw material



Pour the molten material into a cavity



After solidification, take out the object



Car engine block



Water tanks



Plastic toys



Individual turbine blade



Base of heavy machines



Candle



Casing of TV



Sculpture



Videos related to sand mould casting process

https://www.youtube.com/watch?v=XlRK_SMWX7Y

<https://www.youtube.com/watch?v=1oZnxZj6-Ig>

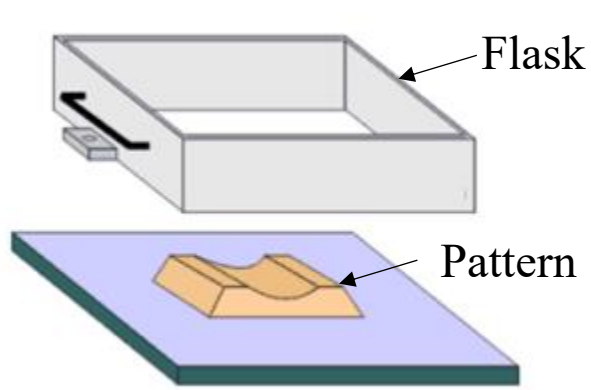
<https://www.youtube.com/watch?v=h2RiLz1-v4Q>

<https://www.youtube.com/watch?v=oOOg4eGLuwg>

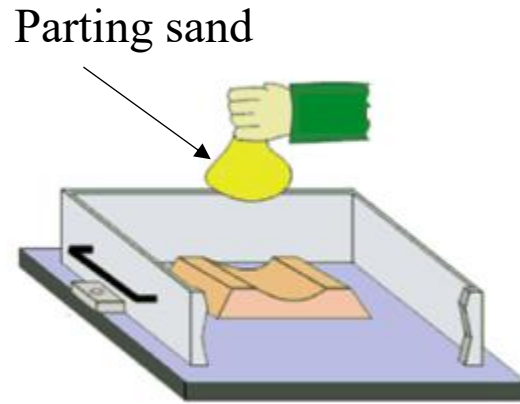
<https://www.youtube.com/watch?v=PkjpmPLNKZs>



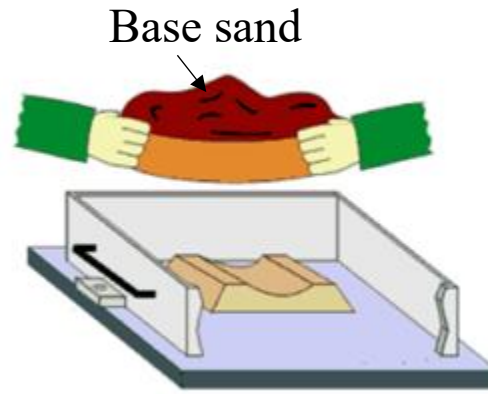
Sand mould casting steps



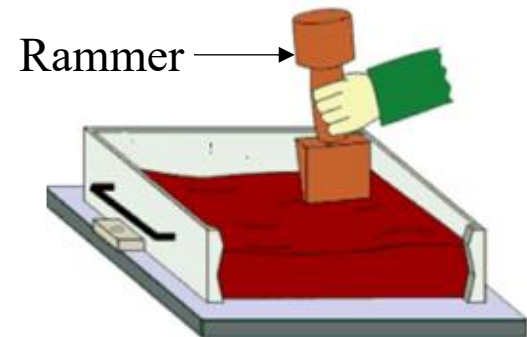
Place one half of a pattern on a base plate and attach the flask



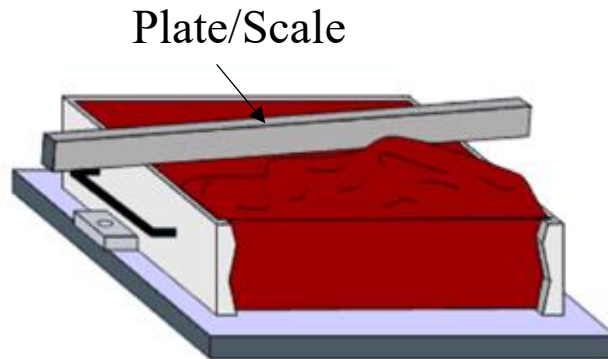
Pour some parting sand to cover the pattern



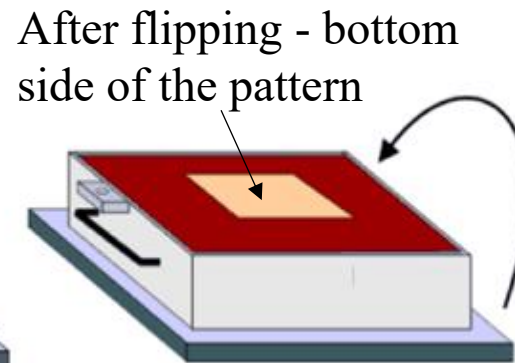
Pour the base sand to partially fill the flask



Heavily press the sand by hand rammer to make the sand tightly packed



Again pour base sand and do ramming. Finally remove excess sand

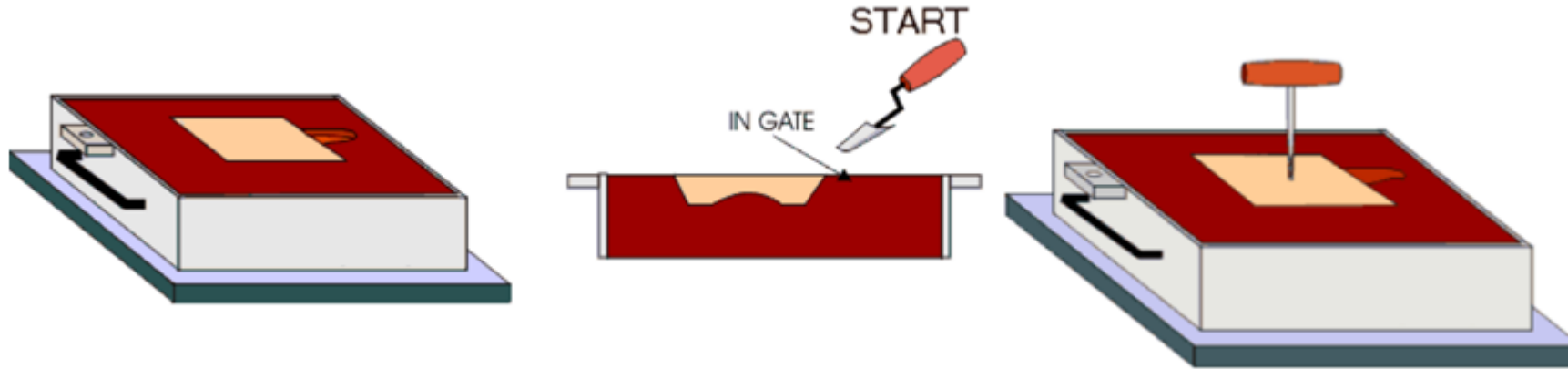


Put another base plate at top and then gently flip the entire arrangement

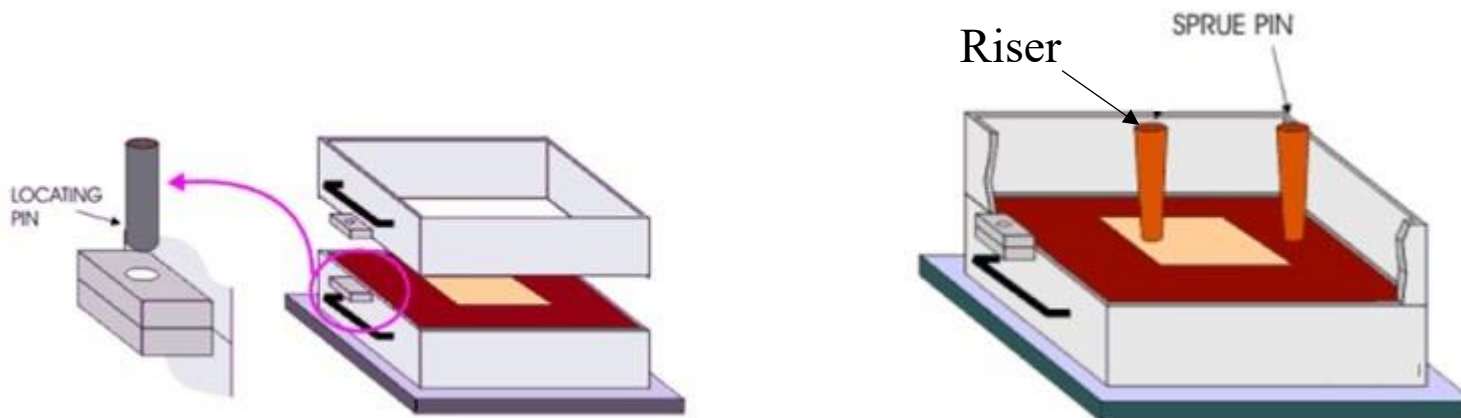




Sand mould casting steps (continued)



Make a small opening (called *in gate*) adjacent to the pattern for molten metal entry

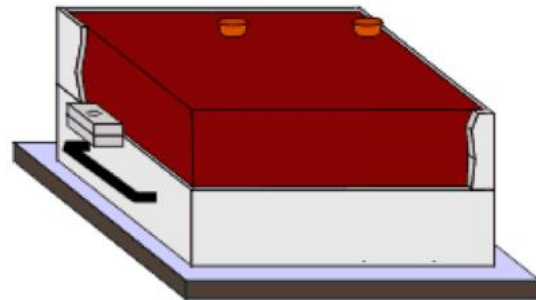


Put the top flask and attach a pin to hold the top flask with bottom one

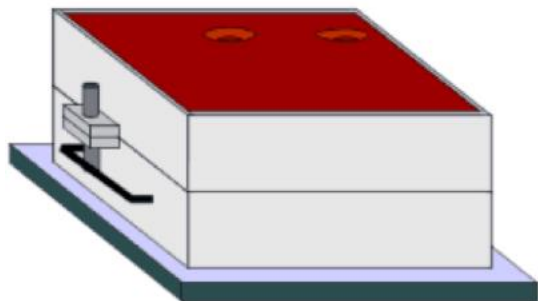
Put two wooden pins – one as riser and another one called sprue pin



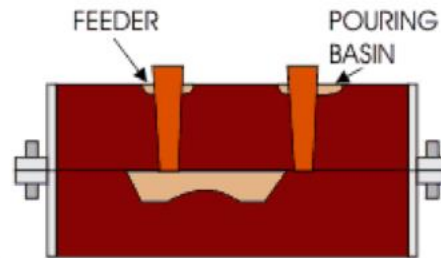
Sand mould casting steps (continued)



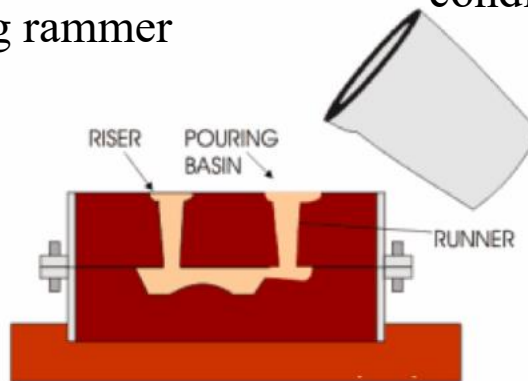
Again pure base sand and press using rammer



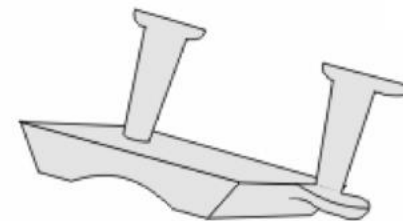
Take out two pins gently.
Isometric view in final condition with two openings



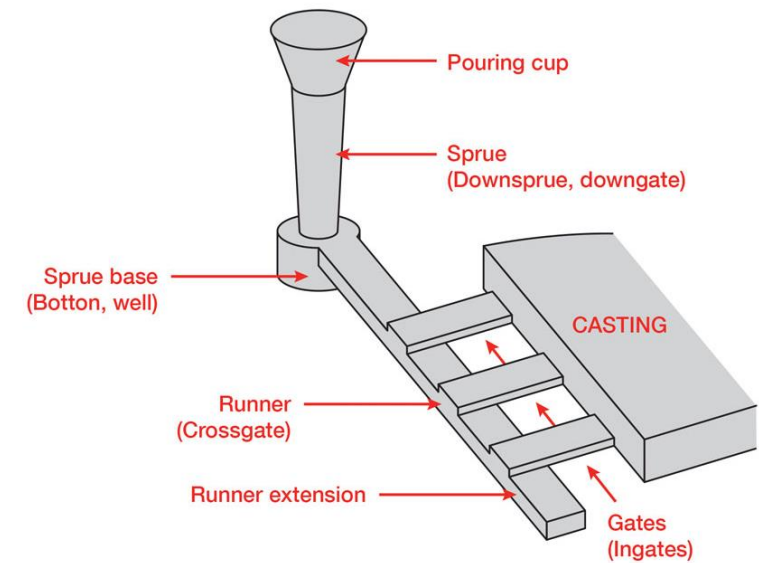
Sectional view at final condition

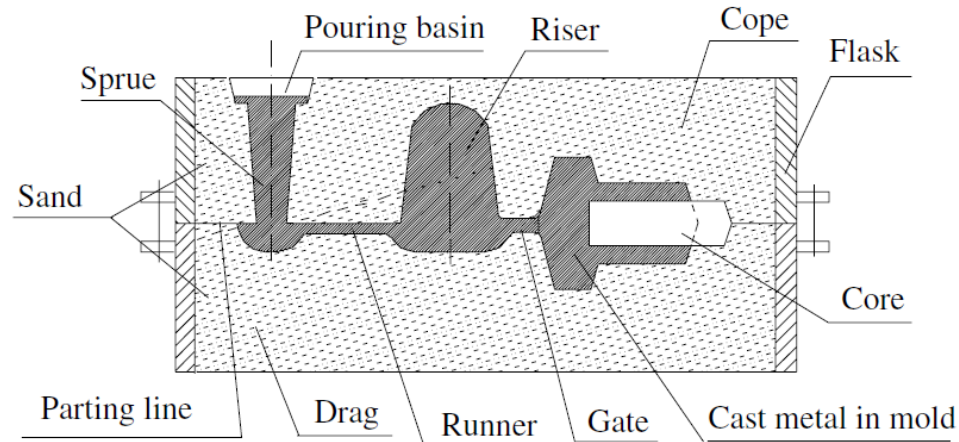


Pour the molten metal through one opening that is connected to the *in gate*



After cooling and solidification, remove the sand and take out the desired object





Flask: An external metal (or sometimes wood) frame in which the mold is formed. Depending upon the position of the flask in the molding structure, it is referred to by various names such as **drag** – lower molding flask, **cope** – upper molding flask, (**cheek** – intermediate molding flask if any).

Parting line: This is the dividing line between the two molding flasks that makes up the mold.

Pattern: It is the replica of the final object to be made. The mold cavity is made with the help of pattern.

Molding sand: Silica sand, which binds strongly without losing its permeability to air or gases. It is a mixture of river bed silica sand, clay, and moisture in appropriate proportions.

Core: An optional separate part of the mold, made of sand, which is used to create openings and various shaped cavities or hollow portions in the castings.

Pouring basin: A small funnel shaped cavity at the top of the mold into which the molten metal is poured.

Sprue: The passage through which the molten metal, from the pouring basin, reaches the mold cavity.

Runner: The channel through which the molten metal is carried from the sprue to the gate.

In Gate: A channel through which the molten metal enters the mold cavity.

Chaplets: Chaplets are used to support the cores inside the mold cavity to take care of its own weight.

Riser: A column of molten metal placed in the mold to feed the castings as it shrinks and solidifies.

Vent: Small opening in the mold to facilitate escape of air and gases.

The five basic steps in sand castings:

1. Pattern making
2. Core making (optional)
3. Molding
4. Melting of metal and pouring into cavity
5. Cooling, mold dismantling, and cleaning the object

Pattern making:

What is the pattern: Replica of the part (either exact or with some modification) to be cast and is used to prepare the mold cavity. It is the physical model of the casting used to make the mold.

Pattern Material: Usually wood. But can be wax, metal, plaster-of-paris, or plastic.

The mold is made by packing some readily formed aggregate material, such as molding sand, surrounding the pattern.

When the pattern is withdrawn, its imprint provides the mold cavity. This cavity is filled with metal to become the casting.



This is the metal object you need



This is the wooden pattern required

Desirable properties of pattern material

- Easily worked, shaped and joined
- Light in weight
- Strong, hard and durable
- Resistant to wear and abrasion
- Resistant to corrosion, and to chemical reactions
- Dimensionally stable and unaffected by variations in temperature and humidity
- Available at low cost



➤ Benefits of wood as pattern material

1. Cheap
2. Easily available,
3. Light weight,
4. Easiness in surfacing,
5. Easiness in preserving (by shellac coating),
6. Easily workable,
7. Ease in joining,
8. Ease in fabrication

➤ Disadvantages of wood as pattern material

1. Absorbs moisture and changes dimension,
2. Wear by sand abrasion,
3. Warp during forming,
4. Not suitable for rough use,
5. Must be properly dried/ seasoned,
6. Should be free from knots.

Common woods: Burma teak, pine wood, mahogany, Sal, Deodar, Shisham, Walnut, Apple tree

Metal as pattern material:

1. Used for mass production
2. For maintaining closer dimensional tolerances on casting.
3. More life when compared to wooden patterns
4. Do not absorb moisture
5. Do not warp, retain their shape
6. Greater resistance to abrasion
7. Accurate and smooth surface finish
8. More expensive

Common metals: Al alloys, Brass, white metals, Mg alloys, Steel, cast Iron for mass production

Plastic as pattern material:

1. Light weight
2. Good formability
3. Do not absorb moisture
4. Good corrosion resistance

Common plastics: Epoxy resins with fillers, PU foam

Plaster-of-Paris as pattern material

1. Complicated shapes can be cast
2. Has high compressive strength
3. Used for making small and intricate pattern and core boxes

Wax is also used in specialized applications such as investment casting. It provides good surface finish and high accuracy